CLINICAL TEACHING NOTE | FAMILY MEDICINE

Dizziness in the Elderly

Assessment and management in primary care

1. Diagnostic approach

Step 1: Classify by timing and trigger (TiTrATE approach)

- **Episodic, triggered (seconds):** provoked by rolling over, lying back or looking up — typically BPPV.
- **Episodic, spontaneous (minutes–hours):** cardiac arrhythmia, TIA, vestibular migraine, panic/anxiety.
- **Continuous vestibular syndrome (days):** vestibular neuritis, labyrinthitis, or posterior circulation stroke.
- Older, symptom-quality descriptors ("vertigo" vs "light-headedness" vs "unsteadiness") are unreliable in elderly patients, who often mix descriptions — timing and triggers are more discriminating.

Step 2: Key history

- **Onset, duration and triggers:** positional change, standing, exertion, meals, head turning.
- **Associated symptoms:** palpitations, chest pain, syncope, hearing loss, tinnitus, headache, diplopia, dysarthria, limb weakness.
- **Medication review:** antihypertensives, diuretics, nitrates, sedatives/hypnotics, antidepressants, dopaminergic drugs, alcohol — dizziness is often multifactorial and drug-related in older people.
- **Falls history:** frequency, injuries, fear of falling, and impact on independence.
- **Cardiovascular history:** known arrhythmia, valve disease, ischaemic heart disease, pacemaker.
- Visual and hearing impairment, peripheral neuropathy, and cognitive status, which all compound imbalance.

Step 3: Examination

- **Orthostatic BP:** lying, then standing at 1 and 3 minutes — a fall of ≥20 mmHg systolic or ≥10 mmHg diastolic (or symptomatic drop) confirms orthostatic hypotension.
- **Cardiovascular exam:** pulse rate and rhythm, murmurs, signs of heart failure.
- **HINTS exam** (head impulse, nystagmus, test of skew) in acute continuous vertigo — helps separate peripheral (vestibular neuritis) from central (stroke) causes; a normal head impulse test with acute vertigo is a red flag for stroke.
- **Dix–Hallpike manoeuvre** for suspected BPPV (positional, brief episodes).
- Gait, Romberg, vision and hearing testing, and a falls-risk/postural stability assessment.

2. Distinguishing the common causes

	BPPV	Orthostatic hypotension	Cardiac arrhythmia	Vestibular neuritis / stroke
Timing	Seconds, positional	On standing, seconds–minutes	Sudden, seconds, unpredictable	Continuous, days
Trigger	Rolling in bed, looking up	Standing, especially after meals or heat	None, or exertion/palpitation	None; worse with head movement

	BPPV	Orthostatic hypotension	Cardiac arrhythmia	Vestibular neuritis / stroke
Symmetry / exam	Positive Dix–Hallpike	BP drop ≥20/10 mmHg on standing	Irregular or slow/fast pulse	Abnormal HINTS; ask re central signs
Other features	Nausea, brief spinning	Pre-syncope, blurred vision, pallor	Palpitations, chest pain, syncope	Hearing loss/tinnitus (neuritis) or ataxia, dysarthria (stroke)
Course	Recurs with position; responds to Epley	Fluctuates with drugs, hydration, meals	Depends on rhythm; needs cardiology input	Neuritis improves over days–weeks; stroke needs urgent imaging

Other contributors to keep in mind

- **Polypharmacy and multisensory deficit:** combined visual, vestibular, proprioceptive and drug effects producing chronic unsteadiness without a single cause.
- **Vestibular migraine:** episodic vertigo with headache, photophobia or prior migraine history.
- **Carotid sinus syndrome:** unexplained falls or syncope, especially with neck turning or collar pressure.
- Anaemia, hypoglycaemia, hypothyroidism and electrolyte disturbance as reversible metabolic contributors.

Terminology

Vertigo: an illusion of movement of self or surroundings.

Presyncope: a feeling of impending faint, often with pallor and light-headedness.

Disequilibrium: unsteadiness on standing or walking without a spinning sensation.

3. Investigations

Most causes are identified on history and examination. A reasonable primary-care set:

- **Orthostatic BP measurement** (lying and standing at 1 and 3 minutes) in every elderly patient presenting with dizziness or falls.
- **ECG** for rhythm and conduction abnormalities; 24–48-hour Holter monitoring if an intermittent arrhythmia is suspected.
- **Bloods:** FBC, glucose, U&E, calcium, TSH, and B12 if peripheral neuropathy or gait disturbance is present.
- **Dix–Hallpike testing** to confirm BPPV when the history is positional.
- **Audiometry** if hearing loss or tinnitus accompanies vertigo (Ménière's disease, acoustic neuroma).
- **Carotid sinus massage** (with monitoring) if carotid sinus syndrome is suspected in unexplained falls.
- **MRI brain** if the HINTS exam or other findings suggest a central cause, or if there is rapid progression or focal neurology.
- **Medication review** as a routine part of the work-up, not only when polypharmacy is suspected.

Red flags: same-day or emergency referral

- Sudden, severe headache ("thunderclap") or new neurological deficit.
- Abnormal HINTS exam in acute vertigo (normal head impulse test, direction-changing nystagmus, or skew deviation) — suggests posterior circulation stroke.
- Chest pain, palpitations, or syncope with an abnormal ECG.
- Sudden unilateral hearing loss with vertigo.

- New ataxia, dysarthria, diplopia or limb weakness.
- Recurrent falls with injury, or head injury on anticoagulants.

4. Management

General measures for all causes

- **Medication review and deprescribing:** stop or reduce drugs that lower blood pressure, sedate, or impair balance where possible.
- **Correct vision and hearing:** updated glasses prescription, cataract review, hearing aid assessment.
- **Home safety and falls prevention:** remove trip hazards, adequate lighting, grab rails, footwear review, and a falls-prevention exercise programme (e.g. strength and balance training).
- **Hydration and gradual position change:** sit on the edge of the bed before standing; avoid rapid postural change.
- **Explain the diagnosis:** many older patients fear stroke or a serious cardiac event; clear explanation of a benign cause (e.g. BPPV) reduces anxiety and improves adherence to treatment.

BPPV

- **Epley (canalith repositioning) manoeuvre** is first-line and highly effective; can be taught for home use (Brandt–Daroff exercises) if recurrent.
- **Vestibular suppressants** (e.g. cinnarizine, prochlorperazine) are not routinely needed and can worsen central compensation — reserve for severe acute nausea only, for the shortest possible course.
- Refer to ENT or audiovestibular medicine if the manoeuvre fails or the diagnosis is uncertain.

Orthostatic (postural) hypotension

Treat only if symptomatic; asymptomatic postural drops usually need review of contributing drugs rather than new treatment.

Line	Drug	Typical dose	Key cautions
First line	**Non-drug measures**	Slow position changes, leg crossing/muscle pumping, compression stockings, head-up tilt of the bed, adequate hydration and salt (if not fluid-restricted)	Review and reduce causative drugs first: diuretics, alpha-blockers, antihypertensives, antidepressants, dopaminergic agents.
First line	**Fludrocortisone**	0.1–0.3 mg/day	Fluid retention, hypokalaemia, heart failure; monitor potassium and weight.
Second line	**Midodrine**	2.5–10 mg, 2–3 times/day (not before bed)	Supine hypertension, piloerection, urinary retention; avoid in significant heart disease.
Second line	**Pyridostigmine**	30–60 mg, 2–3 times/day	GI cramps, diarrhoea, sialorrhoea; useful when supine hypertension limits other agents.

Cardiac arrhythmia

- **Refer to cardiology** for further rhythm monitoring and consideration of pacing or rate/rhythm control.
- Correct reversible contributors: electrolyte disturbance, thyroid dysfunction, and review of QT-prolonging or negatively chronotropic drugs.

Vestibular neuritis

- **Short course of vestibular suppressants** (3–5 days maximum) for acute symptom relief, then stop — prolonged use delays central compensation.
- **Vestibular rehabilitation exercises** promote recovery and are particularly beneficial in older patients with slower compensation.

Suspected central cause (posterior circulation stroke/TIA)

- **Refer immediately for emergency assessment and imaging** — do not manage in primary care.

5. Follow-up

- Review symptom response, falls, and postural BP at **2–4 weeks** after any medication change or new diagnosis.
- Reassess drug list at every review — dose changes elsewhere (e.g. a new antihypertensive) can re-trigger orthostatic symptoms.
- Revisit the diagnosis if new features develop; a first presentation labelled as BPPV or orthostatic hypotension can evolve or coexist with another cause.
- Consider a formal falls-risk assessment or referral to a falls clinic for patients with recurrent dizziness and falls.

Key take-home points

- Classify by timing and trigger, not by how the patient describes the sensation.
- Always check orthostatic BP and review the drug chart — dizziness in the elderly is often multifactorial and iatrogenic.
- Use the HINTS exam to separate a benign peripheral cause from a dangerous central one in acute vertigo.
- BPPV responds well to the Epley manoeuvre; avoid long-term vestibular suppressants.
- Falls prevention (vision, footwear, home safety, strength and balance training) is part of management for every cause.

References

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This summary is an educational aid compiled from standard family medicine and geriatric-medicine references; it is not a substitute for local clinical guidelines or the primary literature.